

2g. Content of Additional Pure Mathematics (Optional paper Y545)

Introduction to Additional Pure Mathematics.

In **Additional Pure Mathematics** learners will broaden and deepen their knowledge of pure mathematics, studying both discrete and continuous topics which form the foundation of undergraduate study in mathematics and mathematical disciplines. This area covers recurrence relations, number theory, group theory, the vector product, surfaces and partial differentiation.

8.01 Sequences and Series

Recurrence relations are explored, including their long term behaviour and solution of first and second order recurrence relations.

8.02 Number Theory

Number theory is introduced through number bases, modular arithmetic, divisibility algorithms and solving linear congruences. This foundation is extended to simultaneous linear congruences and Fermat's little theorem.

8.03 Groups

Group axioms and examples of finite groups of small order are studied, including cyclic groups. Lagrange's theorem is applied to the order of subgroups and the concept of an isomorphism is introduced.

8.04 Further Vectors

The concept of vector product introduced in the Pure Core is developed and extended to the scalar triple product, finding areas and volumes, and solving problems using vector methods.

8.05 Surfaces and Partial Differentiation

Surfaces in 3-D are explored, including contours and sections, and applying partial differentiation to find stationary points, and to classify them using the Hessian matrix.

8.06 Further Calculus

The calculus techniques in the Pure Core are extended to finding arc-lengths and surface areas of revolution and the use of reduction formulae.

Assumed knowledge

Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but will only be explicitly assessed where it appears in this section.

Use of technology

To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. **Graphing tools:** Learners could use graphing software to investigate the relationships between graphical and algebraic representations, including 3-D plots of surfaces and solids of revolution.
2. **Computer Algebra System (CAS):** Learners could use CAS software to investigate algebraic relationships, including manipulation of matrices, evaluating integrals, solving recurrence systems and solving equations and as an investigative problem solving tool. This is best done in conjunction with other software such as graphing tools and spreadsheets.
3. **Visualisation:** Learners could use appropriate software to visualise situations in 3-D relating to surfaces, and to linear transformations and invariance.
4. **Spreadsheets:** Learners could use spreadsheet software to investigate sequences and series, for modelling and to generate tables of values for functions.

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When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
8.01 Sequences and Series			
8.01a	Recurrence relations	a) Be able to work with general sequences given either as recurrence relations or by position-to-term (closed form) formulae $u_n = f(n)$. <i>The notation $\{u_n\}$ for sequences, which may or may not include a zeroth term, should be recognised.</i>	
8.01b		b) Use induction to prove results relating to both sequences and series.	
8.01c	Properties of sequences	c) Understand and be able to describe various possibilities for the behaviour of sequences. <i>Learners are expected to be able to use the terms "periodic", "convergence", "divergence", "oscillating", "monotonic".</i> <i>Note that a periodic sequence with period two may be referred to as "oscillating", but that both convergent and divergent sequences can oscillate. "Divergence" can refer to sequences that are bounded or unbounded.</i>	
8.01d		d) Identify and be able to use the limit of the n th term of a sequence as $n \rightarrow \infty$, including steady-states. <i>Includes forming sequences from other sequences, for example, finding differences or ratios of successive terms of a sequence.</i> <i>[Rates of convergence are excluded.]</i>	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
8.01e	Fibonacci and related numbers	e) Be able to work with the Fibonacci numbers (and other Fibonacci-like sequences, such as the Lucas numbers), and understand their properties. <i>Includes recognising and using the properties of ϕ, both numerical and algebraic, and its role in the Fibonacci sequence.</i>	
8.01f 8.01g	Solving recurrence systems	f) Be able to solve a first-order linear recurrence relation with constant coefficients, using the associated auxiliary equation and complementary function. <i>Includes finding both general and particular solutions.</i> <i>Includes homogeneous and non-homogeneous recurrence relations of the form $u_{n+1} = au_n + f(n)$, where $f(n)$ may be a polynomial function or of the form dk^n.</i> <i>Includes knowing the terms, “closed form” and “position-to-term”.</i> <i>Includes understanding that a “recurrence system” consists of a “recurrence relation”, an “initial condition” and the range of the variable n.</i>	g) Be able to solve a second-order linear recurrence relation with constant coefficients, using the associated auxiliary equation and complementary function. <i>Includes finding both general and particular solutions.</i> <i>Includes the cases when the roots of the auxiliary equation are:</i> (i) <i>distinct and real,</i> (ii) <i>repeated,</i> (iii) <i>complex.</i> <i>Includes homogeneous and non-homogeneous recurrence relations of the form $u_{n+2} = au_{n+1} + bu_n + f(n)$, where $f(n)$ may be a polynomial function or of the form dk^n.</i> <i>Learners should be aware that this topic is the discrete analogue of the work on differential equations in H245 section 4.10.</i>
8.01h 8.01i	Modelling	h) Be able to apply their knowledge of recurrence relations to modelling. <i>Includes birth- and/or death-rates and the use of the $\text{INT}(x)$ function for discrete models. Learners may find it useful to have a calculator with this function, but large numbers of repeated applications will not be required in the assessment.</i>	i) Be able to extend their knowledge of modelling to second order recurrence relations.

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